

Data Wrangling

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Armed Forces Frequency Tables I

Description

To turn the pivot table into two data frames, the plan is to first create a table for a group of soldiers and another for individual soldiers to handle the two separate cases. I will also use a pay grade and rank mapping to determine the pay per rank. In the group table, I will include the branch, pay grade, rank, and frequency of soldiers, then merge this data with the pay-rank mapping. For the individual soldier table, I will reference the group table and repeat each row based on the count of soldiers in that rank and position so that each row represents one soldier. Each soldier will have a unique ID so that the data set contains labeled information for every soldier, including their rank, pay grade, and branch.

Data

Google Sheets Armed Forces Data

Packages

‘dplyr’ - for data wrangling and filtering ‘janitor’ - for cleaning column names ‘googlesheets4’ - to read the Data ‘kableExtra’ - to format output tables ‘tidyr’ - for reshaping data

Functions

‘read_sheet()’ - imports the data from Google ‘mutate()’, ‘filter()’, ‘across()’ - for cleaning and manipulating columns ‘pivot_longer()’ / ‘pivot_wider()’ - for reshaping the data between groups ‘kbl()’ 0 displays tables neatly in report

```
group_soldiers
```

```
# A tibble: 336 x 4
  PayGrade Branch Sex Frequency
  <chr>    <chr>    <chr>    <dbl>
1 E1      Air Force Female    1933
2 E1      Air Force Male    8537
3 E2      Air Force Female    2019
4 E2      Air Force Male    7343
5 E3      Air Force Female   10369
6 E3      Air Force Male   37324
7 E4      Air Force Female   15055
8 E4      Air Force Male   53185
9 E5      Air Force Female   10762
10 E5     Air Force Male   40614
# i 326 more rows
```

```
individual_soldiers
```

```
# A tibble: 7,668,972 x 4
  soldier_id PayGrade Branch Sex
  <int> <chr>    <chr>    <chr>
1      1 E1      Air Force Female
2      2 E1      Air Force Female
3      3 E1      Air Force Female
4      4 E1      Air Force Female
5      5 E1      Air Force Female
6      6 E1      Air Force Female
7      7 E1      Air Force Female
8      8 E1      Air Force Female
9      9 E1      Air Force Female
10     10 E1      Air Force Female
# i 7,668,962 more rows
```

Armed Forces Frequency Tables II

Description

The frequency table below summarizes the number and percentage of male and female service members holding the E5 pay grade across all U.S. military branches. For each branch, the table presents the total count of personnel by sex along with the corresponding percentage within that branch. The final row, “All Branches,” combines totals across all branches to show the overall gender distribution for the E5 rank. Overall, the table indicates that males make up the majority of service members at the E5 pay grade in every branch of the military.

```
freq_kable
```

Table 1: Frequency Table: Rank E5 - Male vs Female by Branch

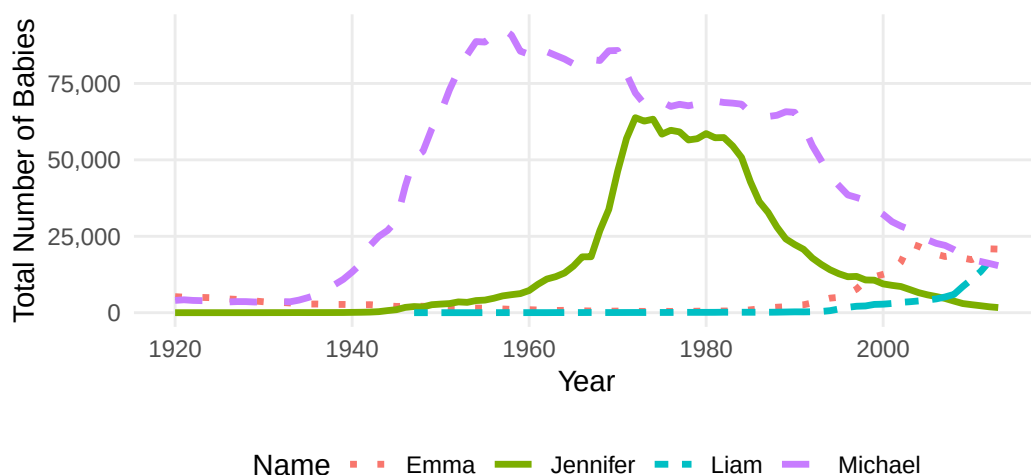
Branch	Sex	Frequency
AirForce	Female	10,762 (20.95%)
AirForce	Male	40,614 (79.05%)
Army	Female	10,954 (16.66%)
Army	Male	54,803 (83.34%)
Marine	Female	2,670 (10.71%)
Marine	Male	22,262 (89.29%)
Navy	Female	16,169 (21.76%)
Navy	Male	58,142 (78.24%)
Space	Female	173 (16.76%)
Space	Male	859 (83.24%)
All Branches	Female	40,728 (18.73%)
All Branches	Male	176,680 (81.27%)

Popularity of Baby Names

When examining the popularity of the selected baby names Jennifer, Michael, Emma, and Liam there were noticeable differences in their usage over time. These names were chosen to illustrate a range of naming trends. Jennifer was most popular in the 1960s and 1970s, which is reflected in its peak during that period. Michael has been consistently common for almost a century, representing a name that has remained popular over time. Emma has gradually increased in popularity in recent decades, showing a slow upward trend. Finally, Liam has experienced rapid growth in recent years, demonstrating its emergence as one of the most popular contemporary names.

Popularity of Selected Baby Names Over Time (1920

Jennifer peaked in the 1970s–80s while Liam shows—
recent explosive growth



Data: dcData::BabyNames

Plotting a Mathematical Function

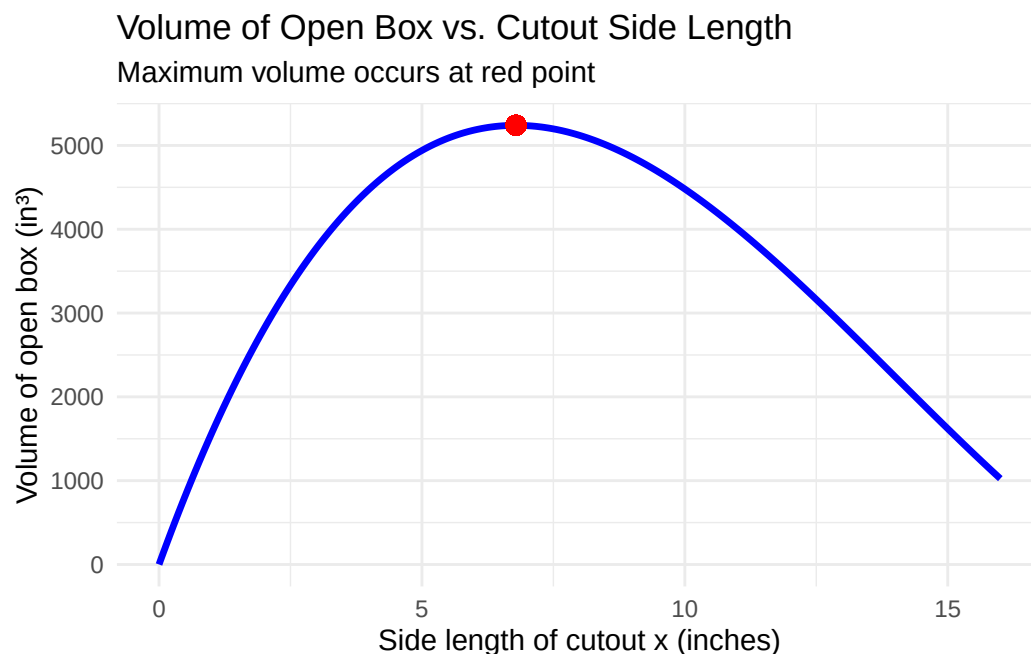
The Function Max_Volume calculates the volume of an open-top box using the formula:

$V(x) = (L - 2x)(W - 2x)(x)$, where: x = side length of the square cutouts, L = length of the paper, and W = width of the paper

The length and width of the resulting box decrease by twice the cutout size, and the height of the box is x

The plot shows that the volume of the box rises with increasing cutout size, reaches its peak, and then decreases as the base area becomes too small. The maximum volume occurs at a cutout size of about 6.79 inches, giving a box volume of approximately 5239.81 cubic inches. This suggests that cutting squares of around 6.79 inches provides the most efficient box size, achieving the best balance between height and base dimensions.

Plot of cutout sizes



What I've Learned so far

Throughout the length of this course, we have gone over many different topics involving how to wrangle and organize data, the different ways to tidy up data within a data frame, and also various methods for plotting data with a data visualization and how to make that visualization impact and as efficient as possible. The use of data frames within a set of data allows the opportunity to arrange many different variables and organize them into anyway you need in order to represent some pattern or trend, or even help get a solution to an ongoing problem. Tidying and wrangling data allows us to pull information from the internet or from other created data table, and shape

it into something entirely new and find new values and meaning from the original data. The work within Quarto helps us create clean documentation and various types of text markdown files that can formally represent our data and findings in a professional format and help make the information as clear as possible. Data visualization techniques also really showed me how I should go about when designing a plot or collecting data to include the most important aspects and make my visual as appealing and interpret able as possible for the best chance at being understood and used. The use of many different R packages has also shown the usefulness of R when it comes to handling statistical data and plotting/organizing information.

Code Appendix

Armed Forces Frequency Tables I

```
library(dplyr)
library(tidyverse)
library(rvest)
library(readr)
library(google Sheets4)
# get the google sheets URL
sheet1 <- "https://docs.google.com/spreadsheets/d/19xQnI1cB"
sheet2 <- "h6Jkw7eP8YQuuicMlVDF7Gr-nXCb5qbwb_E/export?format=csv"
# merge sections of URL
sheet_url <- paste0(sheet1, sheet2)
# read data
pivot_data <- read_csv(sheet_url, header = FALSE, stringsAsFactors = FALSE)
# extract branch names/sex from row 2 and 3
header_branches <- as.character(pivot_data[2, ])
header_sexes <- as.character(pivot_data[3, ])
# fill in missing branch names
for (i in 2:length(header_branches)) {
  if (is.na(header_branches[i]) || header_branches[i] == "") {
    header_branches[i] <- header_branches[i - 1]
  }
}
combined_headers <- paste(header_branches, header_sexes, sep = "_")
combined_headers[1] <- "PayGrade"
colnames(pivot_data) <- combined_headers
pivot_data <- pivot_data[-c(1,2,3), ]
group_soldiers <- pivot_data %>%
  # convert wide to long format
  pivot_longer(
    cols = -PayGrade, # ignore PayGrade column
    names_to = c("Branch", "Sex"), # split up branch and sex columns
    names_sep = "_",
    values_to = "Frequency" # new frequency column name
```

```

) %>%
# filter out empty values and total column from gender
filter(!is.na(Frequency) & Frequency != "") %>%
filter(Sex != "Total") %>%
#convert frequency from string to numeric
mutate(Frequency = as.numeric(gsub(",", "", Frequency))) %>%
# sort by branch, paygrade, sex
arrange(Branch, PayGrade, Sex)
#individual soldier data frame
individual_soldiers <- group_soldiers %>%
  filter(!is.na(Frequency) & Frequency > 0) %>%
  # make sure no nonzero/empty frequencies
  uncount(Frequency) %>%
  #expand each row by frequency
  mutate(soldier_id = row_number()) %>%
  # soldier_id for soldiers
  select(soldier_id, everything()) %>%
  # order soldier_id in front
  arrange(Branch, PayGrade, Sex)

```

Armed Forces Frequency Tables II

```

library(dplyr)
library(knitr)
library(kableExtra)
library(google sheets4)
library(tidyr)

# Load the pivot table data
sheet1 <- "https://docs.google.com/spreadsheets/d/19xQnI1cB"
sheet2 <- "h6Jkw7eP8YQuuicMlVDF7Gr-nXCb5qbwb_E/export?format=csv"
sheet_url <- paste0(sheet1, sheet2)
pivot_data <- read.csv(sheet_url, header = FALSE, stringsAsFactors = FALSE)

# Assign column names
colnames(pivot_data) <- c(
  "PayGrade",
  "Army_Male", "Army_Female", "Army_Total",
  "Navy_Male", "Navy_Female", "Navy_Total",
  "Marine_Male", "Marine_Female", "Marine_Total",
  "AirForce_Male", "AirForce_Female", "AirForce_Total",
  "Space_Male", "Space_Female", "Space_Total",
  "Total_Male", "Total_Female", "Total_Total"
)

```

```

# Clean data
pivot_data <- pivot_data %>%
  mutate(across(-PayGrade, ~ as.numeric(gsub(",", "", .)))) %>%
  slice(-(1:3)) %>%
  filter(!(PayGrade %in% c("Total Enlisted", "Total Warrant Officers",
                          "Total Officers", "Total"))) %>%
  slice(-n())

# Rank to check
rank_to_check <- "E5"

# Filter for that rank
rank_data <- pivot_data %>%
  filter(grepl(rank_to_check, PayGrade, ignore.case = TRUE)) %>%
  select(PayGrade,
         Army_Male, Army_Female,
         Navy_Male, Navy_Female,
         Marine_Male, Marine_Female,
         AirForce_Male, AirForce_Female,
         Space_Male, Space_Female)

# Reshape into long format
rank_long <- rank_data %>%
  pivot_longer(
    cols = -PayGrade,
    names_to = c("Branch", "Sex"),
    names_sep = "_",
    values_to = "Count"
  )

# Frequency table with merged Frequency column
freq_table <- rank_long %>%
  group_by(Branch) %>%
  mutate(Total_Branch = sum(Count, na.rm = TRUE),
         Percent = round((Count / Total_Branch) * 100, 2),
         Frequency = paste0(formatC(Count, format = "d", big.mark = ","),
                             " (", Percent, "%)")) %>%
  ungroup() %>%
  select(Branch, Sex, Frequency) %>%
  arrange(Branch, Sex)

# Add total row at bottom
total_row <- rank_long %>%
  group_by(Sex) %>%
  summarise(Count = sum(Count, na.rm = TRUE)) %>%
  mutate(Branch = "All Branches") %>%
  group_by() %>%

```

```

mutate(Percent = round((Count / sum(Count)) * 100, 2),
       Frequency = paste0(formatC(Count, format = "d", big.mark = ","),
" (", Percent, "%)") %>%
select(Branch, Sex, Frequency)

freq_table <- bind_rows(freq_table, total_row)

# Display table nicely
freq_kable <- kbl(freq_table,
                  caption = paste("Frequency Table: Rank", rank_to_check,
"- Male vs Female by Branch"),
                  align = "lcc",
                  booktabs = TRUE) %>%
kable_styling(full_width = FALSE, position = "center", font_size = 12) %>%
row_spec(0, bold = TRUE)

```

Popularity of Baby names

```

library(dcData)
library(dplyr)
library(ggplot2)
data(BabyNames)
#load baby names dataset

#four names selected
selected_names <- c("Jennifer", "Michael", "Emma", "Liam")
names_filtered <- BabyNames %>%
  filter(name %in% selected_names)
# aggregate counts by name and year
names_by_year <- names_filtered %>%
  group_by(name, year) %>%
  summarise(total_count = sum(count), .groups = 'drop')
# filter to include years from 1920 and on
names_by_year <- names_by_year %>%
  filter(year >= 1920)
# line plot showing popularity of names over time
ggplot(names_by_year, aes(x = year, y = total_count, color = name,
linetype = name)) +
  geom_line(linewidth = 1.2) +
  # assign different line style to each name
  scale_linetype_manual(values = c("Jennifer" = "solid",
"Michael" = "dashed",
"Emma" = "dotted",
"Liam" = "twodash")) +
#format y axis with commas

```



```

scale_y_continuous(labels = scales::comma_format()) +
theme_minimal() +
theme(
  plot.title = element_text(face = "bold", size = 14),
  plot.subtitle = element_text(size = 10, color = "gray30"),
  legend.position = "bottom",
  panel.grid.minor = element_blank()
) +
# add labels, title and caption
labs(
  title = "Popularity of Selected Baby Names Over Time (1920-2015)",
  subtitle = "Jennifer peaked in the 1970s-80s while Liam shows-
recent explosive growth",
  x = "Year",
  y = "Total Number of Babies",
  color = "Name",
  linetype = "Name",
  caption = "Data: dcData::BabyNames"
)

```

Plotting a Mathematical Function

```

# equation calculation to return volume of cutout box
Max_Volume <- function(x, paper_length = 36, paper_width = 48){
  volume <- (paper_length - (2*x))*(paper_width - (2*x))*x
  return(volume)
}

```

```

library(ggplot2)
# sequence of x value from 0 to 4.25
x_values <- seq(0, 16, by = 0.01)
# data frame with x values and volumes
volume_df <- data.frame(
  x = x_values,
  volume = Max_Volume(x_values)
)
# plot showing relationship between cutout size and volume
ggplot(volume_df, aes(x = x, y = volume)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(aes(x = x_values[which.max(volume_df$volume)],
    y = max(volume_df$volume)),
    color = "red", size = 3) + # highlight maximum
#labels and title
labs(
  title = "Volume of Open Box vs. Cutout Side Length",

```

```
    subtitle = "Maximum volume occurs at red point",  
    x = "Side length of cutout x (inches)",  
    y = "Volume of open box (in³)"  
  ) +  
  # simple theme  
  theme_minimal()
```